

The spectral index of Fermi acceleration at ultra-relativistic shocks: 25 digits, all dimensions, arbitrary scattering, and the breakdown of the spectrum–anisotropy identity

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(Dated: October 3, 2026)

Test particles that diffuse isotropically in direction angle at an ultra-relativistic shock acquire a power-law spectrum $f \propto p^{-s}$. For a 3D gas, published values of the limiting index span the range 4.22–4.24. The most precise is 4.227 ± 0.001 , and an analytic approximation gives $38/9$. We reduce the $\Gamma_u \rightarrow \infty$ problem to a single Legendre operator with two weights, joined across the shock by a power-law factor. In this limit the upstream eigenfunctions are exact Laguerre (3D) or Hermite (2D) functions. We then solve the eigenfunction-matching condition in ball arithmetic with up to 320 modes. A model-selected extrapolation, whose truncation error contains an $N^{-6} \log N$ term, gives $s_\infty = 4.226\,978\,816\,696\,084\,932\,553\,273$ in 3D ($\beta_d = 1/3$) and $s_\infty = 3.298\,508\,056\,390\,718\,143\,338\,299$ in 2D ($\beta_d = 1/2$; energy index $2.2985\dots$). Both are confirmed by an independent discrete-ordinates solver and by a finite- Γ solver. The numbers exclude $38/9$ in 3D and $(1 + \sqrt{13})/2$ for the 2D energy index. They also show that the Keshet–Waxman spectrum–anisotropy relation, presented as exact for isotropic diffusion, is violated: the grazing-incidence anisotropy is $0.85950(2)$ against the required 0.83512 (3D), and $0.70965(1)$ against 0.66532 (2D). We trace the violation to a non-analytic corner of the forward–backward transport problem at grazing incidence on the shock. We further give the exact return probability $P_{\text{ret}} = 0.378985$ (3D), in conflict with the Monte-Carlo value 0.437 ± 0.002 , the full curve $s_\infty(\beta_d)$ to ~ 18 digits, its slope 2.879467949325874 at $\beta_d \rightarrow 0$, and the leading finite- Γ correction.

The method extends to three further settings. (i) In d spatial dimensions the upstream modes are generalized Laguerre functions $L_n^{((d-3)/2)}$. The energy index decreases monotonically from 2.2985 ($d = 2$) to the Newtonian strong-shock value: numerically $s_E(d) = 2 + 2/d - 7.1(1) d^{-3/2} + \dots$, with the limit 2 confirmed to 3×10^{-8} . (ii) Upstream scattering drops out of the $\Gamma_u \rightarrow \infty$ limit entirely. For downstream scattering we give the exact linear-response kernel $K(\mu) = \delta s / \delta \ln D(\mu)$. It is smooth and shows no enhanced sensitivity at grazing incidence, and it reproduces nonperturbative results for weak anisotropy. For the tangled-field scattering law of Kirk et al. we obtain $s_\infty = 4.2126801838$, a shift of -0.0143 (approximations: -0.02 and $+0.03$). (iii) For a strong shock into a cold gas with the Jüttner–Synge equation of state, we compute s at all speeds. It falls *below* the Newtonian value 4 for $\Gamma_u \beta_u < 0.9249$, reaching a minimum $s = 3.988970$ at $\Gamma_u \beta_u = 0.5541$.

I. INTRODUCTION

First-order Fermi acceleration at relativistic collisionless shocks is the standard explanation for the non-thermal electrons that radiate in gamma-ray-burst afterglows, relativistic jets and pulsar-wind nebulae. It is also the process studied in particle-in-cell (PIC) simulations of Weibel-mediated shocks [1–3]. The canonical theoretical benchmark is the test-particle problem: an infinitely thin planar shock moving with Lorentz factor $\Gamma_u \rightarrow \infty$ into an unmagnetized medium, with particles diffusing isotropically in direction angle on both sides. Because the problem has no intrinsic momentum scale, the spectrum is a power law $f \propto p^{-s}$, and s is a pure number fixed by kinematics and the downstream equation of state.

That number has been computed many times:

- Monte-Carlo simulations gave $s \approx 4.2$ [4] and 4.22 ± 0.01 [5].
- The eigenfunction method of Kirk et al. (KGGA) [6], which builds on [7, 8], gave 4.23 ± 0.01 .
- A moment expansion gave 4.24 at order six [9].

- A finite-difference relaxation code gave 4.227 ± 0.001 [10].

- Keshet and Waxman (KW) [11] derived an identity between s and the anisotropy of particles moving along the shock front, presented as exact for isotropic diffusion. Truncating it gave the closed form $s = (3\beta_u - 2\beta_u\beta_d^2 + \beta_d^3)/(\beta_u - \beta_d)$, hence $s = 38/9$ in the ultra-relativistic limit.

In 2D, which is the geometry of most PIC studies, the analogous approximation gives an energy index $s_E = (1 + \sqrt{13})/2$ [10, 12].

In this paper we solve the benchmark problem to 25 significant digits in both 3D and 2D, and establish the following:

1. The index (Sec. V). The values exclude the closed forms above and settle the spread among numerical values.
2. The spectrum–anisotropy identity is not exact. It fails because the shock-front distribution is non-analytic at grazing incidence (Sec. VI).

3. The exact return probability per cycle disagrees with the published Monte-Carlo value (Sec. VII).

We also give the full dependence $s_\infty(\beta_d)$ on the downstream speed, its expansion about the infinitely compressive limit, and the leading finite- Γ correction. Three extensions follow:

- the index in any spatial dimension, and its approach to the Newtonian value (Sec. VIII);
- the exact response of s to an arbitrary scattering law, including KGGA's tangled-field model (Sec. IX);
- the index for physical strong shocks at all speeds (Sec. X).

The computations used the BootLoops harness [13]: Arb ball arithmetic [14] and its integer-relation gate [15].

II. FORMULATION

We follow KGGA [6]. In each fluid's rest frame the particle has momentum p and direction cosine μ with respect to the shock normal. The distance z is measured in the shock frame; the flow is along $+z$, with the upstream at $z < 0$ moving at speed β_u and the downstream at $z > 0$ moving at speed β_d . The stationary transport equation is

$$\Gamma(\beta + \mu) \partial_z f = \partial_\mu [(1 - \mu^2) D \partial_\mu f]. \quad (1)$$

Here D is constant on each side (isotropic diffusion in direction angle), and Γ and β take their upstream or downstream values. The boundary conditions are $f \rightarrow 0$ as $z \rightarrow -\infty$ and f bounded as $z \rightarrow +\infty$. The invariant f is continuous at $z = 0$ under the boost $\beta_r = (\beta_u - \beta_d)/(1 - \beta_u \beta_d)$:

$$p_+ = \Gamma_r p_- (1 + \beta_r \mu_-), \quad \mu_+ = \frac{\mu_- + \beta_r}{1 + \beta_r \mu_-}, \quad (2)$$

where $-$ and $+$ label the upstream and downstream frames. Separating variables gives modes $Q(\mu) e^{\Lambda z/\Gamma}$ with

$$((1 - \mu^2) Q')' = \Lambda(\beta + \mu) Q, \quad (3)$$

with Q regular at $\mu = \pm 1$. Downstream, the allowed modes have $\Lambda \leq 0$; this includes the isotropic mode $\Lambda = 0$, which carries particles to $z = +\infty$. Upstream, the allowed modes have $\Lambda > 0$. The equation is of forward-backward type: the weight $\beta + \mu$ changes sign at the grazing direction $\mu = -\beta$, where particles move parallel to the shock in the shock frame.

A. The limit $\Gamma_u \rightarrow \infty$

Let $\varepsilon = 1 - \beta_u$ and $y = (1 + \mu_-)/\varepsilon$. Then $\beta_u + \mu_- = \varepsilon(y - 1)$, $1 - \mu_-^2 = \varepsilon y(2 - \varepsilon y)$, and $\partial_\mu = \varepsilon^{-1} \partial_y$, so to leading order Eq. (3) upstream becomes

$$\partial_y (2y \partial_y g) = \tilde{\Lambda} (y - 1) g. \quad (4)$$

Substituting $g = e^{-ay} h$ with $2a^2 = \tilde{\Lambda}$ gives Laguerre's equation in $2ay$ with index $n = (a - 1)/2$. Solutions regular at $y = 0$ and bounded as $y \rightarrow \infty$ therefore exist exactly for $a = 2n + 1$:

$$Q_n(y) = e^{-(2n+1)y} L_n((4n+2)y), \quad \tilde{\Lambda}_n = 2(2n+1)^2. \quad (5)$$

These are the asymptotic eigenfunctions of Kirk & Schneider [6, 16], here obtained in closed form. Define $\kappa = (1 + \beta_d)/(1 - \beta_d)$, so $\kappa = 2$ for $\beta_d = 1/3$. Then $1 - \beta_r = \kappa \varepsilon + O(\varepsilon^2)$, and Eq. (2) becomes

$$\mu_+ = \frac{y - \kappa}{y + \kappa}, \quad \frac{p_+}{p_-} = \sqrt{\frac{\varepsilon}{2\kappa}} (y + \kappa). \quad (6)$$

Write $F(\mu_+)$ for the downstream-frame angular distribution at the shock, defined by $f = p_+^{-s} F$. It is related to the upstream one, $g(y)$, by

$$F(\mu_+) \propto (y + \kappa)^s g(y) \propto (1 - \mu_+)^{-s} g. \quad (7)$$

The grazing directions coincide: $\mu_+ = -\beta_d$ if and only if $y = 1$.

a. Unified form. Substituting $y = \kappa(1 + \mu)/(1 - \mu)$ gives $\partial_y (2y \partial_y) = \frac{(1-\mu)^2}{2\kappa} \partial_\mu (1 - \mu^2) \partial_\mu$ and $y - 1 = (\kappa + 1)(\beta_d + \mu)/(1 - \mu)$. In the downstream variable both sides therefore obey the same operator, with different weights:

$$((1 - \mu^2) Q')' = \Lambda w Q, \quad w = \begin{cases} \beta_d + \mu & (\text{down}), \\ \frac{\beta_d + \mu}{(1 - \mu)^3} & (\text{up}). \end{cases} \quad (8)$$

The two sides are glued by Eq. (7). The upstream weight has an irregular singular point at $\mu = 1$, which produces the essential decay $e^{-(2n+1)y}$.

B. Two dimensions

For momenta confined to a plane with angle ϕ to the normal, isotropic angular diffusion replaces the right side of Eq. (3) by ∂_ϕ^2 , giving $Q'' = \Lambda(\beta + \cos \phi) Q$ with Q even. Near $\phi = \pi$, set $\theta = \pi - \phi = (2\varepsilon)^{1/2} t$; the upstream problem becomes $Q_{tt} = \lambda(t^2 - 1) Q$, a harmonic oscillator. Its even, decaying solutions with $y = t^2 = (1 + \mu_-)/\varepsilon$ are

$$Q_n(y) = e^{-(4n+1)y/2} L_n^{(-1/2)}((4n+1)y), \quad \lambda_n = (4n+1)^2. \quad (9)$$

The kinematics, Eq. (7), are unchanged. We take $\beta_d = 1/2$, the 2D analogue of $\beta_d = 1/3$ (an ultra-relativistic 2D gas has adiabatic index $3/2$).

III. METHOD

a. Petrov–Galerkin condition. F lies in the closed span of the downstream $\Lambda \leq 0$ modes if and only if it is orthogonal, in the indefinite inner product $\langle a, b \rangle = \int (\beta_d + \mu) a b d\mu$, to every downstream mode with $\Lambda > 0$. This follows from the full-range orthogonality and completeness of the eigenfunctions of forward–backward equations [6, 17]. Expand $g = \sum_{n < N} a_n Q_n$ and test against the N lowest downstream $\Lambda > 0$ modes Q_j^+ (in 2D the measure is $d\phi$). The condition becomes $\det S(s) = 0$ with

$$S_{jn}(s) = \int_{-1}^1 (\beta_d + \mu) (y(\mu) + \kappa)^s Q_n(y(\mu)) Q_j^+(\mu) d\mu. \quad (10)$$

This is KGGA’s “new method” [6] in the limit, with exact trial functions.¹

b. Downstream eigenpairs. In the orthonormal Legendre basis $\{p_k\}$, Eq. (3) becomes the three-term recurrence

$$k(k+1)c_k + \Lambda(\beta c_k + b_{k-1}c_{k-1} + b_k c_{k+1}) = 0, \quad (11)$$

with $b_k = (k+1)/\sqrt{(2k+1)(2k+3)}$. In 2D the cosine basis gives k^2 in place of $k(k+1)$, with $b_0 = 2^{-1/2}$ and $b_{k \geq 1} = 1/2$. Regularity at both ends selects the minimal solution, computed by backward recurrence. The eigenvalues are the roots of $\beta + b_0 c_1/c_0 = 0$. They are seeded in double precision from the reduced symmetric tridiagonal problem and refined by secant iteration.

c. Arithmetic and quadrature. All quantities are Arb balls [14] at $200 + 3N$ bits. The precision must grow with N because the trial basis (5) loses about 0.73 digits of conditioning per mode. The integral (10) is split at $\mu = -\beta_d$. On $[-1, -\beta_d]$ we use Gauss–Legendre with $8N + 120$ nodes; on $y \in [1, 2^9]$ we use dyadic Gauss–Legendre panels with 80 nodes each. The Legendre series are truncated at $6N + 120$ terms. The root in s is found by Newton iteration with $d \log \det S/ds = \text{tr}(S^{-1}S')$.

d. Extrapolation. Table I lists s_N for N up to 320 (3D) and 160 (2D); Fig. 1 shows the convergence. The local exponent tends to 4 from above. To choose the asymptotic model, we fit each candidate exactly on $m+1$ consecutive values of N and predict the next, larger N out of sample. The model

$$s_N = s_\infty + aN^{-4} + bN^{-5} + cN^{-6} \log N + \sum_{k \geq 6} d_k N^{-k} \quad (12)$$

beats pure integer powers, powers of $N^{-1/3}$ and $N^{-2/3}$, and log terms at other orders. Its out-of-sample errors

TABLE I. Truncated eigenvalues s_N (digits shown are exact for each N).

N	3D ($\beta_d = 1/3$)	2D ($\beta_d = 1/2$)
2	4.227011365546354259	3.298517209472822393
4	4.226979904553625690	3.298508343664055128
8	4.226978864665305610	3.298508069242913564
16	4.226978819232631672	3.298508057071648065
32	4.226978816842089231	3.298508056429920311
64	4.226978816704844244	3.298508056393069977
128	4.226978816696621322	3.298508056390862158
256	4.226978816696118116	
320	4.226978816696098497	
∞	4.226978816696084933	3.298508056390718143

TABLE II. Model selection for the 3D extrapolation ($N \leq 320$). “pred. err.” is the out-of-sample error when predicting the largest N from the next $m+1$ smaller values; $s_\infty - 4.2269788166960849325532$ is in units of 10^{-26} .

model	m	pred. err.	s_∞ offset
integer powers $N^{-4, -5, \dots}$	6	6.6×10^{-25}	34.3
	8	1.0×10^{-25}	60.2
powers of $N^{-1/3}$ from N^{-4}	8	1.2×10^{-26}	79.2
Eq. (12)	6	1.3×10^{-26}	72.1
	7	2.9×10^{-27}	72.6
	8	8.2×10^{-28}	72.8

are about 100 times smaller (Table II), and its estimate of s_∞ changes by less than 2×10^{-27} between fit orders $m = 7$ and 8.

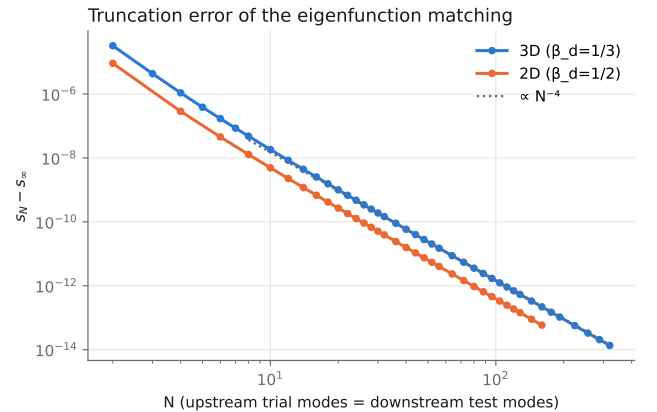


FIG. 1. Truncation error $s_N - s_\infty$ of the eigenfunction matching in 3D and 2D.

¹ In y -space the integrand is $(y-1)(y+\kappa)^{s-3} Q_n Q_j^+$. Ref. [6] prints the factor as $(y+2)^{-s}$. Our finite- Γ solver (Sec. IV), which uses the exact boost Eq. (2), converges to Eq. (10).

IV. VALIDATION

1. *Discretization.* At $N = 128$, the value of s_N is unchanged to 50 digits when we increase the y -panel nodes ($80 \rightarrow 200$), the μ -nodes ($1144 \rightarrow 1600$), the Legendre truncation ($888 \rightarrow 1100$), the y cutoff, or the working precision. The out-of-sample residuals of Eq. (12), $< 10^{-27}$, bound any discretization noise at large N .
2. *Independent method.* We also apply discrete ordinates to the unified form (8). Both sides are collocated on the same n Gauss–Legendre nodes, where the operator is exact on polynomials. The allowed subspaces, of dimensions $n_>$ and $n_<$ (nodes on either side of $-\beta_d$), sum to n . The index then solves $\det[V_d | \text{diag}((1 - \mu_k)^{-s}) V_u] = 0$. This method uses no Laguerre functions, no quadrature of Eq. (10), and no projection. Its convergence is oscillatory, but it agrees with s_∞ to $\sim 10^{-11}$ for $n \leq 256$.
3. *The limit itself.* A double-precision finite- Γ solver computes the upstream modes from the full Legendre pencil at speed $\beta_u < 1$ and uses the exact boost. At fixed N it converges to the limiting s_N as Γ_u^{-2} ; extrapolation reproduces the limit to 2×10^{-7} .
4. *Physical controls.* The finite- Γ solver reproduces nonrelativistic diffusive shock acceleration, $s \rightarrow 3r/(r-1)$ with corrections $O(\beta_u^2)$, for compression ratios $r = 4$ and 2.5 . In the limit solver, $s_\infty \rightarrow 3$ as $\beta_d \rightarrow 0$, where every particle returns.

V. RESULTS

A. The universal index

The 3D index is

$$s_\infty^{3D} = 4.226\,978\,816\,696\,084\,932\,553\,273 \quad (13)$$

for $\beta_d = 1/3$. The uncertainty is below one unit in the last digit, estimated from the spread of the best-ranked extrapolations. The 2D index is

$$s_\infty^{2D} = 3.298\,508\,056\,390\,718\,143\,338\,299 \quad (14)$$

for $\beta_d = 1/2$, corresponding to the energy index $s_E = s - 1 = 2.298\,508\dots$. Figure 2 compares the 3D value with the literature. It confirms 4.227 ± 0.001 [10] and the cruder values. It excludes 38/9 (off by 4.76×10^{-3}). In 2D it excludes $s_E = (1 + \sqrt{13})/2$ (off by 4.27×10^{-3}).

a. Closed form? We used BootLoops’ `pslq_gate`, which requires two-precision stability and planted positive controls inside each basket. It finds no relation among $\{1, s, s^2, s^3\}$ with coefficients $\leq 10^5$. It finds none expressing s as a rational combination of

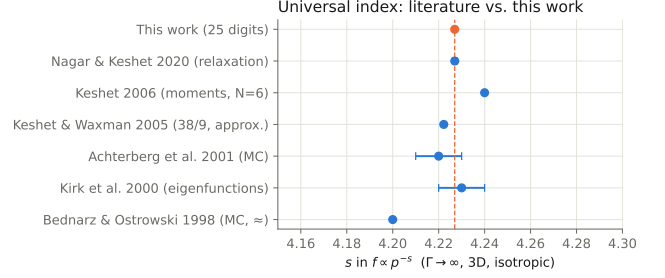


FIG. 2. Published values of the 3D index [4–6, 9–11] compared with this work.

TABLE III. $s_\infty(\beta_d)$, 3D, $\Gamma_u \rightarrow \infty$ (estimated error $\lesssim 10^{-17}$), and the KW formula.

β_d	s_∞	KW
1/1000	3.002881218520578911	3.003001
1/100	3.028970806963360501	3.030102
1/10	3.306754917747891770	3.312222
1/5	3.658025754515797887	3.660000
1/4	3.855120826049169944	3.854167
3/10	4.070689184355443892	4.067143
2/5	4.578744853592199682	4.573333
1/2	5.252200597122419889	5.250000
3/5	6.237339907758071001	6.240000
2/3	7.217625960002580662	7.222222
3/4	9.182870847014065742	9.187500
4/5	11.15638644157611112	11.16000
17/20	14.45859724585625469	14.46083

$\{1, \pi, \pi^2, \log 2, \log 3\}$ with coefficients $\leq 10^3$, or of $\{1, \sqrt{2}, \sqrt{3}, \zeta(3)\}$ with coefficients $\leq 10^4$. This holds for both s_∞^{3D} and s_∞^{2D} . At 25 digits these exclusions are necessarily weak.

B. Dependence on the downstream speed

Figure 3 and Table III give $s_\infty(\beta_d)$ for $0 < \beta_d \leq 0.85$, each point to $\sim 10^{-18}$. This curve gives the $\Gamma_u \rightarrow \infty$ index for any downstream equation of state, for example a lower compression caused by magnetization [6], within the same scattering model. The KW formula is accurate to $\leq 5.5 \times 10^{-3}$, but its error changes sign twice, near $\beta_d \simeq 0.23$ and 0.55 . Near the infinitely compressive limit,

$$s_\infty = 3 + a_1 \beta_d + a_2 \beta_d^2 + a_3 \beta_d^3 + \dots, \quad (15)$$

with $a_1 = 2.879\,467\,949\,325\,874$, $a_2 = 1.749\,393\,501\,5$ and $a_3 = 1.176\,62$. The KW formula instead gives $a_1 = 3$.

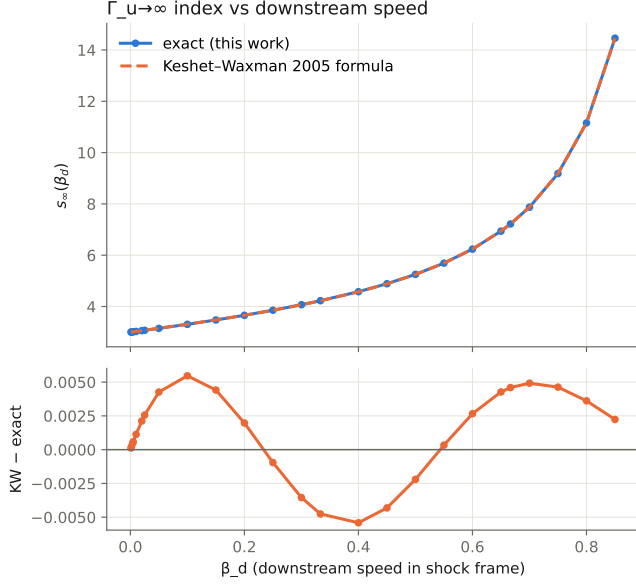


FIG. 3. $s_\infty(\beta_d)$ for $\Gamma_u \rightarrow \infty$ (top) and the error of the Keshet-Waxman formula (bottom).

C. Finite shock Lorentz factor

At fixed $\beta_d = 1/3$, the finite- Γ solver gives

$$s(\Gamma_u) = s_\infty + \frac{0.990}{\Gamma_u^2} + \frac{1.09}{\Gamma_u^4} + \dots, \quad (16)$$

a shift of 0.2% at $\Gamma_u = 10$. For a physical equation of state, β_d itself depends on Γ_u , and the term $(ds_\infty/d\beta_d)\delta\beta_d$ adds to this.

VI. THE SPECTRUM-ANISOTROPY IDENTITY

KW [11] expand the distribution in each fluid frame about grazing incidence, $f_i = [a_0^{(i)} + a_1^{(i)}x + a_2^{(i)}x^2 + \dots]p_i^{-s}$ with $x = \mu_i + \beta_i$. At $x = 0$, Eq. (1) gives $a_2 = -a_1\beta\gamma^2$ for every $z \neq 0$. Taking $z \rightarrow 0$ on both sides and matching a_0 , a_1 and a_2 through Eq. (2) yields one relation between s and the downstream anisotropy $\rho_d = a_1^{(d)}/a_0^{(d)} = F'(-\beta_d)/F(-\beta_d)$. Repeating their algebra symbolically for $\beta_u \rightarrow 1$ reproduces their result in 3D. It also gives the 2D analogue, in which the operator ∂_ϕ^2 gives $a_2 = -a_1\beta\gamma^2/2$:

$$\rho_d^{3D} = \frac{(1 - \beta_d)s - (1 + \beta_d)}{2(1 - \beta_d^2)}, \quad (17)$$

$$\rho_d^{2D} = \frac{s[(1 - \beta_d)s - \beta_d]}{(1 - \beta_d^2)(2s + 1)}. \quad (18)$$

With the exact indices, the identity requires $\rho_d = 0.835117$ (3D) and 0.665318 (2D).

We compute $F'(-\beta_d)/F(-\beta_d)$ from the Galerkin null vector and check it against finite differences. It converges as $N^{-2/3}$ (Fig. 4), and all extrapolation ansätze give

$$\rho_d^{3D} = 0.85950 \pm 0.00002, \quad \rho_d^{2D} = 0.70965 \pm 0.00001. \quad (19)$$

These are violations of 2.9% and 6.7%. Inserting the true anisotropies into Eqs. (17) and (18) would predict $s = 4.292$ and 3.438 . The discrete-ordinates solution independently gives $\rho_d^{3D} \simeq 0.862$ at $n = 48$, on the same decreasing trend. So the identity is not exact, even though KW's closed-form approximation (which assumed $a_1^{(d)}/a_0^{(d)} = \beta_u - \beta_d/2 = 5/6$) happens to land within 5×10^{-3} of the true s .

a. Mechanism. The derivation assumes that the second-order coefficient of the trace $F(\mu)$ equals the limit of $a_2(z)$ as $z \rightarrow 0^\pm$. Near $(z, x) = (0, 0)$, both sides of the unified problem (8) reduce, after rescaling z separately on each side, to the same forward-backward equation $x \partial_\zeta f = \partial_x^2 f$. Write $g = (1 - \mu)^s \tilde{F}$ upstream so that $\tilde{F}$ is continuous across the shock. The two sides then differ only at sub-leading order, through a drift term $(-2s(1 - \beta_d)\partial_x$ upstream) and the $O(x)$ variation of the upstream weight. Under the scaling $x \sim \ell$, $\zeta \sim \ell^3$, this mismatch is smaller by one power of ℓ . It excites a corner solution of degree two, $\zeta^{2/3}\Phi(x/\zeta^{1/3})$. Such a solution leaves a_1 continuous but gives a finite, side-dependent $a_2(0^\pm)$, different from the one-sided second derivatives of the trace, which may contain $x^2 \log|x|$. Three observations agree with this picture:

- the $N^{-2/3}$ convergence of F' , since the upstream modes resolve the grazing point on the Airy scale $N^{-2/3}$;
- the $N^{-4} = (N^{-2/3})^6$ convergence of s_N ;
- the $N^{-6} \log N$ term selected in Eq. (12).

A rigorous corner analysis is left for future work.

VII. RETURN PROBABILITY

Downstream scattering conserves $|p|$ in the downstream frame. A particle entering the downstream at momentum p_+ therefore leaves, if it returns, at the same p_+ . The return probability per cycle is then exactly the ratio of number fluxes through the shock at fixed p_+ :

$$P_{\text{ret}} = \frac{\int_{-1}^{-\beta_d} |\beta_d + \mu| F d\mu}{\int_{-\beta_d}^1 (\beta_d + \mu) F d\mu}. \quad (20)$$

In 2D the measure is $d\phi$. We obtain $P_{\text{ret}} = 0.3789847$ (3D) and 0.3339262 (2D), converged to the digits shown; discrete ordinates give 0.37899 . Ref. [5] reports $P_{\text{ret}} = 0.437 \pm 0.002$ for the same model, measured by counting each particle's downstream-to-upstream crossings until escape, which is the same definition.

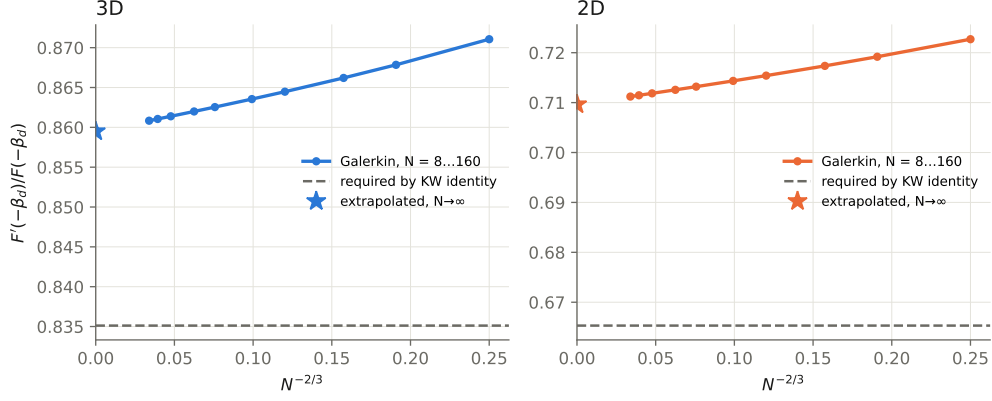


FIG. 4. Grazing-incidence anisotropy $F'(-\beta_d)/F(-\beta_d)$ against $N^{-2/3}$; stars mark the extrapolated limits. The dashed lines are the values required by Eqs. (17) and (18).

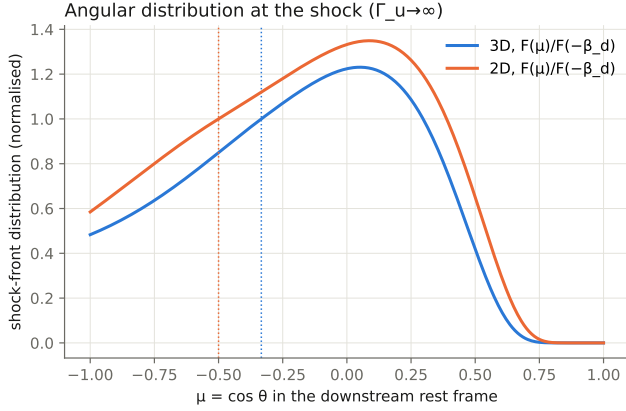


FIG. 5. Downstream-frame angular distribution at the shock, normalized at grazing incidence (dotted lines).

To arbitrate, we simulated the downstream half-space directly. Particles were injected with the flux-weighted entering distribution $(\beta_d + \mu)F(\mu)$, $\mu > -\beta_d$. The direction vector performed Brownian motion on the unit sphere with generator $D\Delta_{S^2}$, and the distance from the shock evolved as $\dot{\zeta} = \mu + \beta_d$, with an absorbing boundary at $\zeta = 12$. With 2×10^5 particles per run, we find $P_{\text{ret}} = 0.3772 \pm 0.0011$ ($D dt = 4 \times 10^{-3}$) and 0.3761 ± 0.0011 (2×10^{-3}), and the exit-angle flux histogram matches $|\beta_d + \mu|F$. This excludes 0.437. A residual 2.9σ shortfall relative to the exact value remains at this statistics. Its sign is the one expected from end-of-step crossing detection, which misses brief returns of grazing particles.

The spectral index of Ref. [5] agrees with ours, so we suspect the discrepancy lies in the cycle bookkeeping. We note also that the check $s = d + \ln(1/P_{\text{ret}})/\ln\langle E_f/E_i \rangle$ used there is approximate, where d is the dimension. The exact steady-state condition is $P_{\text{ret}}\langle g^{d-s} \rangle = 1$ over the distribution of energy gains g per cycle. With the ex-

act P_{ret} , the effective gain implied by the approximate relation is 2.2051 (3D) and 2.3273 (2D).

VIII. ARBITRARY SPATIAL DIMENSION

Isotropic diffusion on S^{d-1} acts on axisymmetric distributions through

$$\mathcal{L}_d = (1 - \mu^2)^{-\frac{d-3}{2}} \partial_\mu (1 - \mu^2)^{\frac{d-1}{2}} \partial_\mu, \quad (21)$$

which reduces to the operators of Sec. II for $d = 3$ and $d = 2$. Its eigenfunctions are the Gegenbauer polynomials $C_k^{(\lambda)}$, with $\lambda = (d-2)/2$ and eigenvalues $-k(k+d-2)$. They are orthogonal with respect to $(1 - \mu^2)^{\lambda-1/2} d\mu$. In the orthonormal basis the downstream recurrence has $k(k+d-2)$ in place of $k(k+1)$, with

$$b_0 = \frac{1}{\sqrt{2(\lambda+1)}}, \quad b_k = \sqrt{\frac{(k+1)(k+2\lambda)}{4(k+\lambda)(k+\lambda+1)}}. \quad (22)$$

In the upstream limit $\mathcal{L}_d$ becomes $2y\partial_y^2 + (d-1)\partial_y$. Repeating the reduction of Sec. II A gives Laguerre's equation with parameter $\alpha = (d-3)/2$. The decaying upstream modes are

$$Q_n(y) = e^{-a_n y} L_n^{(\frac{d-3}{2})}(2a_n y), \quad a_n = 2n + \frac{d-1}{2}, \quad (23)$$

which contains Eq. (5) ($d = 3$) and the Hermite functions ($d = 2$) as special cases. The kinematics, Eq. (7), do not depend on d .

For an ultra-relativistic d -dimensional gas, $p = e/d$, so $\beta_d = 1/d$. We quote the energy index $s_E = s - (d-1)$. For comparison, a Newtonian strong shock in a monatomic d -dimensional gas has compression $r = d+1$. Its index is then $s = dr/(r-1) = d+1$, i.e. $s_E = 2$ in every dimension.

The general- d solver reproduces the $d = 2$ and $d = 3$ results to all 40 digits computed. At large d the

TABLE IV. Energy index $s_E = s - (d - 1)$ for $\Gamma_u \rightarrow \infty$, $\beta_d = 1/d$, with the extrapolation error.

d	s_E	error
2	2.29850805639071814	10^{-18}
3	2.22697881669608492	10^{-17}
4	2.18595312826943077	10^{-16}
5	2.15886819749245212	10^{-16}
6	2.13944467064007694	10^{-16}
8	2.11313234124007319	10^{-15}
10	2.09591143426758012	10^{-15}
16	2.06713513896664777	10^{-14}
32	2.03876545233014850	10^{-13}
64	2.02178264132778	10^{-12}
128	2.0119405751088	10^{-10}
256	2.0064086029238	10^{-10}
512	2.0033806330794	10^{-9}
1024	2.00175910792	10^{-8}

truncation sequence reaches its asymptotic N^{-4} regime only for $N \gtrsim \sqrt{d}$. We therefore scale the N -range as $\max(12, \lceil \sqrt{d} \rceil) \times \{1, \dots, 4\}$; at $d = 1024$ the discretization is converged to 25 digits. Table IV and Fig. 6 give the results. For non-integer d the extrapolation is less clean: the error is 8×10^{-11} at $d = 5/2$, against 10^{-17} at integer d .

a. Large d . We fit $s_E(d)$ for $d \geq 16$ by weighted least squares. A series in $d^{-1/2}$ fits about 10^3 times better than a series in $1/d$ (maximum residual 1.7×10^{-8} versus 1.9×10^{-5}). Leaving the constant free, the constant term converges to 2.00000003 as terms are added; the coefficient of $d^{-1/2}$ is consistent with zero; and the coefficient of $1/d$ converges to 2.000 to within 2×10^{-4} . Hence

$$s_E(d) = 2 + \frac{2}{d} + \frac{c_{3/2}}{d^{3/2}} + O(d^{-2}), \quad c_{3/2} = -7.1 \pm 0.1. \quad (24)$$

The ultra-relativistic index thus approaches the Newtonian strong-shock value from above, with a relativistic excess $2/d$.

A scaling argument shows why the limit is finite and why the series runs in half-integer powers. Set $\mu = x/\sqrt{d}$. Then $\mathcal{L}_d = d(\partial_x^2 - x\partial_x) + O(1)$, an Ornstein–Uhlenbeck operator, and the grazing point $-\beta_d = -1/d$ maps to $x = -d^{-1/2} \rightarrow 0$. Near grazing, $y - 1 \simeq 2x/\sqrt{d}$, so the kinematic factor $(y + \kappa)^s$ with $s \simeq d$ grows like $e^{x\sqrt{d}}$. This cancels exactly against the upstream decay $e^{-a_n y}$ with $a_n \simeq d/2$. What remains is a d -independent limiting problem, perturbed in powers of $d^{-1/2}$. We have not attempted the perturbative computation of $c_1 = 2$ or $c_{3/2}$; it is a natural target.

IX. ARBITRARY SCATTERING

a. Upstream scattering drops out. Let the upstream angular diffusion be $D_u(\mu)$, continuous and positive at $\mu = -1$. With $\mu_- = -1 + \varepsilon y$, the upstream operator becomes $\varepsilon^{-1} \partial_y [2y D_u(-1) \partial_y] + O(1)$. The constant $D_u(-1)$ only rescales z , so the trial functions (5) and (23) are exact for *every* upstream scattering law. KGGa noted that the upstream D matters only near $\mu = -1$ [6]; in the limit it does not matter at all. Hence s_∞ is a functional of the downstream $D(\mu)$ alone.

b. General downstream scattering. For general $D(\mu)$, Eq. (3) becomes $\partial_\mu [(1 - \mu^2) D \partial_\mu Q] = \Lambda(\beta_d + \mu) Q$. We solve it by Galerkin projection in the orthonormal Legendre basis. The stiffness matrix $\int (1 - \mu^2) D p'_k p'_l d\mu$ is computed by Gauss–Legendre quadrature. The mass matrix $\beta_d + J$ is exact and tridiagonal. We eliminate c_0 and compute the reduced pencil’s eigenvectors at high precision in Arb. The isotropic case is reproduced to all digits. For $D = (\mu^2 + 0.01)^{-1/2}$, doubling the Legendre size from 256 to 400 changes s_{16} only in the 28th digit.

c. Linear-response kernel. About isotropic scattering ($d = 3$, $\beta_d = 1/3$), define

$$\delta s_\infty = \int_{-1}^1 K(\mu) \delta \ln D(\mu) d\mu. \quad (25)$$

We compute K by central differences, using Gaussian bumps of width $\sigma = 0.03$ at 61 centres; the spacing is 0.01 near grazing incidence. Three checks support the result:

- *Convergence in N .* At four test points, including grazing incidence, the kernel at $N = 24$ agrees with $N = 16$ to 1.5×10^{-6} .
- *Scale invariance.* Rescaling D uniformly leaves s unchanged, so $\int K d\mu = 0$. The computed kernel satisfies this to 10^{-4} , the level of its smoothing and grid error.
- *Nonperturbative test.* Applied to the tangled-field family below, $\int K \delta \ln D$ reproduces the exact shifts to 0.5% ($a = 10$) and 0.1% ($a = 3$). The deviation then grows smoothly with the anisotropy: 0.9%, 5% and 17% for $a = 2, 1$ and 0.5.

The kernel (Fig. 7) is smooth and broad. It is negative for $\mu \lesssim 0.05$ (minimum -0.32 at $\mu \simeq -0.40$) and positive above (maximum $+0.39$ at $\mu \simeq 0.44$). So faster scattering of particles moving back toward the shock hardens the spectrum, and faster scattering of particles receding downstream softens it. Nothing special happens at grazing incidence: K varies by less than 10^{-2} across $\mu = -1/3 \pm 0.02$, with second differences consistent with a smooth function. A point-like dependence on $D'(-\beta_d)$ would add $c \delta'(\mu + \beta_d)$ to K . Fitting the smoothed kernel on $-0.48 < \mu < -0.18$ with a polynomial background of degree 3–5 plus this template gives c consistent with zero

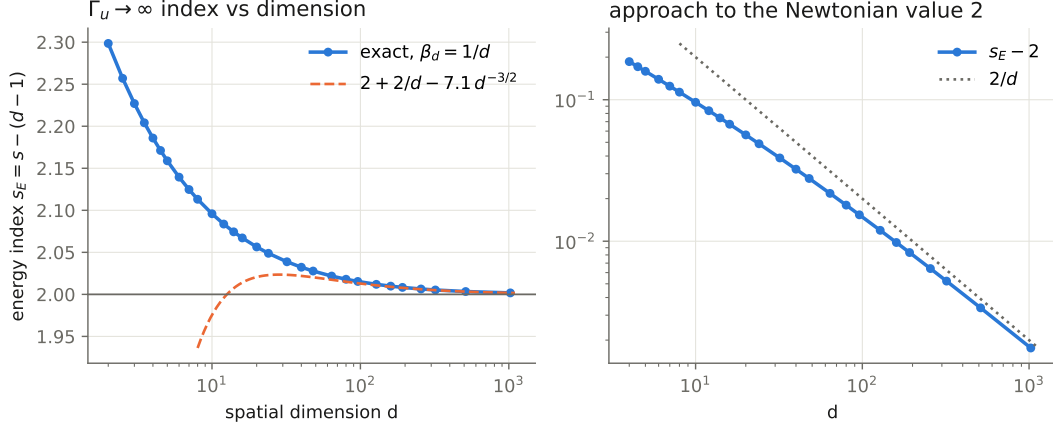


FIG. 6. Left: energy index versus dimension, with the large- d form (24). Right: $s_E - 2$ against $2/d$.

TABLE V. s_∞ for $D \propto (\mu^2 + a^2)^{-1/2}$ downstream (any upstream D), 3D, $\beta_d = 1/3$.

a	s_∞	error
0.05	4.2115053403	3×10^{-11}
0.1	4.2126801838	3×10^{-11}
0.2	4.21536621378	5×10^{-12}
0.3	4.217831856689	6×10^{-13}
0.5	4.22136017681089	3×10^{-14}
1	4.22493766179950	5×10^{-15}
2	4.22640358332686	2×10^{-15}
3	4.22671717421733	1×10^{-15}
10	4.22695486792644	1×10^{-15}
∞	4.22697881669608	(isotropic)

at the 10^{-8} level, so $|c| < 10^{-7}$. This contradicts the expectation, drawn from the spectrum–anisotropy relation [11], that s is particularly sensitive to the derivative of D for particles moving along the shock front.

d. Tangled fields. For KGGA’s model of a field tangled on short scales along the normal, $D \propto (\mu^2 + a^2)^{-1/2}$ [6], Table V gives $s_\infty(a)$. At $a = 0.1$, the case KGGA studied, $s_\infty = 4.2126801838$, a shift of -0.0143 from isotropic. KGGA estimated a decrease of “about 0.02”. KW’s approximation for anisotropic diffusion gives 4.26 [11], which has the wrong sign. The index decreases monotonically as $a \rightarrow 0$, to 4.2115 at $a = 0.05$.

X. PHYSICAL STRONG SHOCKS AT ALL SPEEDS

Consider a strong shock into a cold gas ($w_u = 1$, $p_u = 0$) whose downstream state is a Jüttner–Synge gas at temperature $\theta = kT/mc^2$, with specific enthalpy $w = K_3(1/\theta)/K_2(1/\theta)$ and $p/\rho = \theta$. The jump conditions

close in closed form:

- energy per particle in the downstream frame equals the relative Lorentz factor, $\Gamma_r = w - \theta$;
- the Taub adiabat, $w^2 - 1 = \theta(R + w)$, gives the compression $R = \rho_d/\rho_u = (w^2 - 1 - \theta w)/\theta$;
- mass flux, $\Gamma_u \beta_u = R \Gamma_d \beta_d$, then fixes (β_u, β_d) .

These reproduce $R \rightarrow 4$ nonrelativistically, and $R \rightarrow 4\Gamma_r$ with $\beta_d \rightarrow 1/3$ in the ultra-relativistic limit.

We compute s with the finite- Γ solver, extrapolating $N \in \{8, 12, 16, 20\}$. The accuracy is about 10^{-9} for $\Gamma_u \beta_u \leq 10$, degrading to 10^{-7} and 10^{-6} at $\Gamma_u \beta_u = 19$ and 35 because of double precision. Table VI and Fig. 8 give the results. The index is not monotonic. It falls below the Newtonian value 4 and reaches a minimum

$$s_{\min} = 3.988970 \quad \text{at} \quad \Gamma_u \beta_u = 0.5541, \quad (26)$$

where $\beta_u = 0.4847$, $\theta = 0.0518$ and $R = 4.398$. At the minimum, $\beta_d = 0.1250003$, close to but measurably distinct from $1/8$. The index returns to $s = 4$ at $\Gamma_u \beta_u = 0.9249$ ($\beta_u = 0.6790$, $R = 4.995$) and then rises toward s_∞ .

The dip arises from two competing effects. The downstream gas becomes mildly relativistic ($\theta \sim 0.05$), so its compression rises above 4; the nonrelativistic formula $3R/(R - 1)$ would give 3.88 at the minimum. The anisotropy that sets in at these speeds partly offsets the harder spectrum this would produce.

XI. DISCUSSION

The test-particle, isotropic-scattering, $\Gamma_u \rightarrow \infty$ index is a benchmark, not a prediction for any particular shock. Self-generated Weibel turbulence makes scattering anisotropic and energy dependent, and the index depends on the angular form of D [6, 10]. Within the benchmark, however, the results here are definitive:

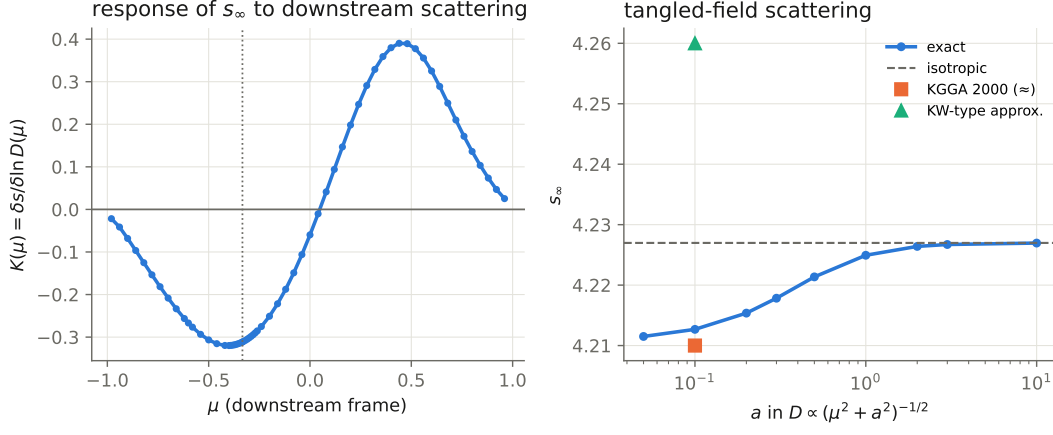


FIG. 7. Left: linear-response kernel of s_∞ to the downstream scattering profile (smoothed on scale $\sigma = 0.03$; the dotted line marks grazing incidence). Right: s_∞ for tangled-field scattering, with the previous estimates at $a = 0.1$.

TABLE VI. Index for a strong shock into cold gas, Jüttner–Synge downstream (3D isotropic scattering).

θ	$\Gamma_u \beta_u$	β_d	R	s
0.001	0.0731	0.01824	4.0075	3.999515259
0.0110	0.2455	0.06002	4.0833	3.995405458
0.0367	0.4595	0.10674	4.2804	3.989734864
0.0518	0.5541	0.12500	4.3984	3.988969566
0.0669	0.6394	0.14011	4.5183	3.989614221
0.122	0.9099	0.18017	4.9678	3.999206186
0.223	1.3423	0.22424	5.8338	4.027235212
0.406	2.0834	0.26639	7.5383	4.076899320
0.740	3.4267	0.29918	10.929	4.136360557
1.35	5.9226	0.31891	17.602	4.183990476
2.46	10.556	0.32810	30.393	4.210297750
4.48	19.088	0.33161	54.305	4.2212949
8.17	34.705	0.33279	98.341	4.225169
∞	∞	1/3	∞	4.226978817

- They fix the reference value to 25 digits.
- They show that two analytic shortcuts (the $38/9$ and $(1 + \sqrt{13})/2$ closed forms, and the spectrum–anisotropy identity behind them) are approximations.
- They identify the grazing-incidence corner of the forward–backward problem as the reason those shortcuts fail and as the factor that controls the convergence of every angular expansion.

The same corner is the likely reason earlier moment and eigenfunction expansions stalled at a few digits. Numerical work that uses the anisotropy identity as a consistency check [11] should treat it as approximate, at the few-percent level in ρ_d . The same machinery gives:

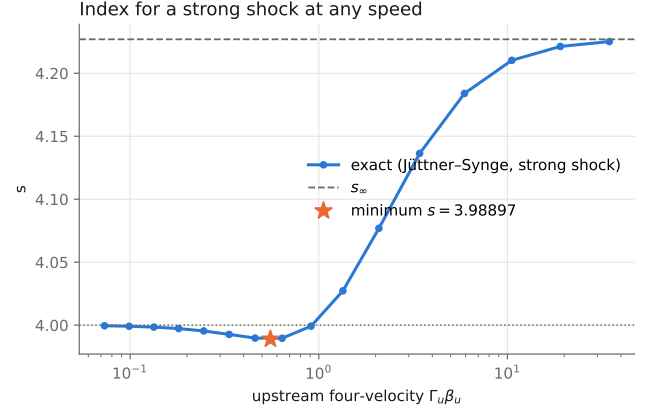


FIG. 8. Spectral index of a strong shock into cold gas (Jüttner–Synge downstream) versus upstream four-velocity. The star marks the sub-Newtonian minimum.

- the index in any dimension, with a numerically established approach $s_E = 2 + 2/d + \dots$ to the Newtonian value;
- a scattering-response kernel that turns a measured downstream $D(\mu)$, e.g. from PIC simulations, directly into a predicted index;
- reference values for physical strong shocks at all speeds, including a sub-Newtonian dip.

Natural next steps are a rigorous local analysis of the corner, a perturbative derivation of the $2/d$ and $d^{-3/2}$ coefficients, and oblique magnetized shocks.

The numerical value rests on an extrapolation in N that is well tested (model selection, independent discretizations, a finite- Γ limit check) but not rigorous. The Arb radii certify each finite- N value, not the limit.

ACKNOWLEDGMENTS

Computations used the BootLoops 1.0 harness [13], python-flint/Arb [14], mpmath [18], NumPy and SciPy, run on a NERSC login node. [Disclosure to be edited by the author: the analysis code and a first draft of this manuscript were produced with the assistance of an AI model (Claude, Anthropic).]

Appendix A: Reproducibility

All code and data are in the accompanying directory: `src/urnd.py` (Galerkin, 3D/2D), `src/dord_hp.py` (discrete ordinates), `src/finite_gamma.py`, `src/model_select.py`, `src/anisotropy.py`, `src/kw_identity.py` (symbolic re-derivation of Eqs. (17)–(18)), `src/return_prob.py`, `src/mc_downstream.py`, `src/scan_beta.py`; for the extensions `src/urnd.py` and `src/scan_d.py` (dimension d), `src/aniso.py`, `src/kernel.py` and `src/scan_aniso.py` (general scattering), and `src/js_curve.py` and `src/js_min.py` (Jüttner-Synge shocks); results in `results/*.json`. A single 3D value s_{64} takes about 6s; s_{320} takes about an hour on one core.

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- [1] A. Spitkovsky, Particle acceleration in relativistic collisionless shocks: Fermi process at last?, *Astrophys. J. Lett.* **682**, L5 (2008).
 - [2] L. Sironi, A. Spitkovsky, and J. Arons, The maximum energy of accelerated particles in relativistic collisionless shocks, *Astrophys. J.* **771**, 54 (2013).
 - [3] L. Sironi, U. Keshet, and M. Lemoine, Relativistic shocks: particle acceleration and magnetization, *Space Sci. Rev.* **191**, 519 (2015), [1506.02034](#).
 - [4] J. Bednarz and M. Ostrowski, Energy spectra of cosmic rays accelerated at ultrarelativistic shock waves, *Phys. Rev. Lett.* **80**, 3911 (1998).
 - [5] A. Achterberg, Y. A. Gallant, J. G. Kirk, and A. W. Guthmann, Particle acceleration by ultrarelativistic shocks: theory and simulations, *Mon. Not. R. Astron. Soc.* **328**, 393 (2001), [astro-ph/0107530](#).
 - [6] J. G. Kirk, A. W. Guthmann, Y. A. Gallant, and A. Achterberg, Particle acceleration at ultrarelativistic shocks: an eigenfunction method, *Astrophys. J.* **542**, 235 (2000), [astro-ph/0005222](#).
 - [7] J. G. Kirk and P. Schneider, On the acceleration of charged particles at relativistic shock fronts, *Astrophys. J.* **315**, 425 (1987).
 - [8] A. F. Heavens and L. O. Drury, Relativistic shocks and particle acceleration, *Mon. Not. R. Astron. Soc.* **235**, 997 (1988).
 - [9] U. Keshet, Analytical study of diffusive relativistic shock acceleration, *Phys. Rev. Lett.* **97**, 221104 (2006), [astro-ph/0608182](#).
 - [10] Y. Nagar and U. Keshet, Diffusive acceleration in relativistic shocks: particle feedback, *Mon. Not. R. Astron. Soc.* **501**, 329 (2021), [1910.10030](#).
 - [11] U. Keshet and E. Waxman, Energy spectrum of particles accelerated in relativistic collisionless shocks, *Phys. Rev. Lett.* **94**, 111102 (2005), [astro-ph/0408489](#).
 - [12] A. Lavi *et al.*, Diffusive shock acceleration in an arbitrary dimension (2020), in preparation; 2D result $s_E \simeq (1 + \sqrt{13})/2$ as quoted in [10].
 - [13] M. D. Schwartz, BootLoops 1.0 (2026), <https://www.bootloops.ai>.
 - [14] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66**, 1281 (2017).
 - [15] H. R. P. Ferguson, D. H. Bailey, and S. Arno, Analysis of PSLQ, an integer relation finding algorithm, *Math. Comp.* **68**, 351 (1999).
 - [16] J. G. Kirk and P. Schneider, *Astron. Astrophys.* **225**, 559 (1989).
 - [17] R. Beals, Partial-range completeness and existence of solutions to two-way diffusion equations, *J. Math. Phys.* **22**, 954 (1981).
 - [18] The mpmath development team, mpmath: a Python library for arbitrary-precision floating-point arithmetic (version 1.3.0) (2023), <https://mpmath.org>.